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How do specific lighting parameters influence the perceived mood and realism of underwater scenes in real-time rendering?

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Graduation Work 2025-2026

Digital Arts and Entertainment

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ABSTRACT & KEY WORDS

Realistic underwater scenes rely on a careful balance between physical light behavior and artistic subjectivity. While modern game engines provide some tools to simulate underwater environments, it might be overwhelming for an indie or beginner developer without experience in this matter, which lighting parameters most strongly influence perceived realism and mood. This graduation work investigates how specific lighting, volumetric parameters and color adjustments affect the perceived visibility, emotional tone, realism, and realism of underwater scenes.

Using Unity HDRP (version 6000.2.6f2), several underwater scenes were created in which volumetric fog attenuation, bloom intensity, particle density, directional light color temperature and color adjustment were manipulated. These parameters were first evaluated in isolation and later combined to represent two real-world Optical Water Types (OWT 2 and OWT 4a). A qualitative user survey was conducted with 33 participants, who evaluated images of the scenes.

The results show that these parameters, specifically volumetric fog attenuation and bloom, have a strong impact on perceived underwater clarity and realism. Light color temperature and color adjustments showed a weak influence on mood perception. However, in scenes that combined other parameters such as volumetric fog and bloom effects, colder lighting contributed to a more tense emotional response. The combined scene results revealed limitations in accurately depicting real-world underwater representations, showing the importance of additional effects such as caustics and god rays.

Overall, this research shows that parameters like volumetric fog attenuation distance, bloom and particles effects, light color temperature and color adjustments have an important role in underwater scene perception. Furthermore, it provides a practical framework to guide developers in better understanding and selecting lighting parameters that balance realism and artistic intent.

Keywords: Underwater Lighting, Volumetric Fog, Light Attenuation, Underwater Post-Processing Effects.

PREFACE

In the scope of the Graduation Work we were asked to choose a research topic relevant to our field of work and delineate a problem, explore existing research, formulate questions, organize experiments, conduct thorough investigations, and propose optimal and innovative solutions.

The broad topic I chose was: Underwater Lighting - Use rendering, lighting and material tricks to create as believable underwater lighting as possible (Light fall off, god rays, caustics, bloom, reflections). The goal was to research how this could be done and find a way to quantify the result of the use-case, so one can compare it to real life shots.

I chose this topic because I was always fascinated with underwater scenes in games, and how they made them so believable and immersive. Although the scope of this research work doesn't allow for a comprehensive study of all aspects that encompass the creation of underwater scenes, it allowed me to better understand the main concepts and the physical behaviors that they try to simulate.

The topic I settled on later into the research work was: How do specific lighting parameters influence the perceived mood and realism of underwater scenes in real-time rendering? This came as a suggestion from the coordinator during one of the first meetings when discussing our topics.

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INTRODUCTION

Recreating realistic underwater scenes is not a new topic and is constantly being expanded on. With the advancement of technology, more powerful tools emerged to better replicate underwater environments with a high level of realism, Fig. 1. Bigger game studios have access to experienced professionals with the knowledge and insight for these tasks, but smaller studios or indie companies might lack that knowledge.

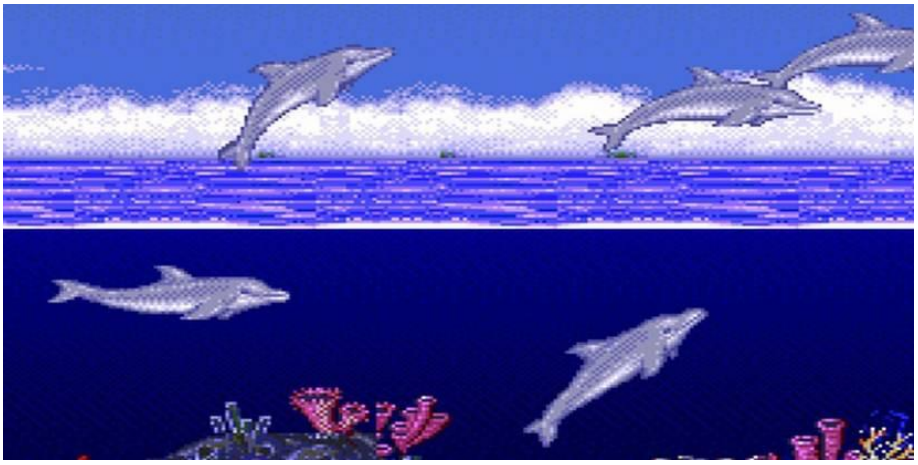

Figure 1 Ecco The Dolphin (1992). Early example of underwater scenes in videogames.

For independent developers or students, implementing underwater rendering can be particularly challenging. One option could be to study and understand the dynamics of water/light interaction on a physical level and then use that knowledge to try and implement it in a Game engine (unreal, unity, etc..), which would also require to understand well the tool those same engines provide. This, however interesting, could be quite challenging and even beyond their current capabilities. The second option would be to search for already made examples of this, for instance, YouTube videos made by developers explaining how they recreate their underwater scenes, which would be less time consuming, as developers would simply recreate what they have done. Yet, if this might not fit the theme/style of the game the developers are trying to make, it would require developers to go back and tweak their process to try and match the goals of the game design. This might be an inefficient process in terms of time and resources. In Fig 2. we see an example of Soma, a survival horror game with a very tense mood and Fig 3. An example of Abzu, an exploration game with beautiful dynamic environments and a more lighthearted mood.

Figure 2 Soma (2015).

Figure 3 Abzu (2016).

Thus, the main idea of this graduation work is to create a simple framework so game developers can identify the right parameters to easily recreate an underwater scene that fits their theme/style goal. This will be done by first doing theoretical research, which will give the reader a basic understanding of the light interactions. Next, insights on what is usually done to simulate underwater environments by established developers will be given, which will be condensed into simple parameters that are the core of simulated underwater environments. Finally, the goal is to orient the user on which parameters have more impact on the overall theme/style goals for their game.

As defined previously, the main research question is *how do specific lighting and volumetric parameters (fog density, color absorption, particle density, and light temperature) influence the perceived mood and plausibility of underwater scenes?* From this we will split the question into simple ones for our hypotheses: *How does volumetric fog affect perceived depth and realism? How does bloom and particle density and distribution influence visibility and immersion? How does directional light temperature and color adjustments modify emotional perception?*

To test the influence of underwater lighting and volumetric parameters on mood and believability, the following hypotheses are proposed:

- H0 – baseline: A simple underwater scene with default directional light only, no volumetric fog or particles. Expected to appear less immersive and neutral in mood. Used as a baseline.
- H1 – Attenuation: Adding volumetric fog simulates light absorption and scattering, enhancing depth perception, reducing contrast, and creating a more believable underwater environment by muting red color and reducing depth, simulating realistic light absorption.[1], [2], [3] We can increase/decrease the fog density and color parameters.
- H2 – Murkiness/haze: The use of bloom and particles effect increases the haze effect and decreasing visibility, and for a more realistic scene, a combination with particle effects and their density can be done. [4], [5], [6]
- H3 – Color Temperature: The adjustment of directional light temperature will modify the scene's tone and influence perceived realism. Colder tones contribute to a more mysterious tone and colder environments, while warmer tones suggest calm or tropical environments. Also, after testing this, some post processing effect will be used to influence the scenes color, like color adjustment, or Tone mapping.[5], [6], [7]

Finally, all previous stated hypotheses will be combined in a combined scene. This should result in a visually cohesive scene with realistic attenuation, subtle murkiness, and mood appropriate color temperature representing the balance between realism and artistic goals.

LITERATURE STUDY / THEORETICAL FRAMEWORK

The literature reviewed here can be categorized into four main topics: (1) Theoretical basis for physical light behavior, (2) Real time and hardware Rendering techniques, (3) Engine specific implementations and (4) Visual/Aesthetic considerations.

1. THEORETICAL BASIS FOR PHYSICAL LIGHT BEHAVIOR

The theoretical basis provides an understanding of light behaviors when it enters another medium, in this case water. There are several main interactions, refraction, reflection, absorption and scattering, Figure 4.[2], [8], [9], [10]

Reflection and refraction occur when light enters another medium and the light rays change direction, Figure 4. Reflection occurs when the light “bounces” on the surface, and refraction occurs when the light “bends”.[8], [9], [10], [11] This phenomena causes a number of effects for the viewer. The caustic effect happens when light is refracted on curved surfaces, such as ripples on the water surface, which creates a visual effect on the surface of objects.[11] Refraction may also cause distortions to the viewer, changing apparent objects size or proximity to the viewer, caused by refraction in the lens of a camera, or swim goggles, where light crossed to another medium, such as air or glass from a camera lens.[9], [10], [11]

While some authors refer to absorption and scattering as attenuations, other call this extinction.[4], [9] Absorption is the loss of energy of the light ray as it passes through another medium, which causes certain wavelengths to be lost, notably reds/oranges, Figure 4 and 5.[2], [7], [8], [10], [11], [12] This gives objects a bluish tone, and the deeper the medium, the less light travels, until, no light can penetrate further, Figure 5.[13] Scattering is the change in direction of the light that occurs when the light ray contacts with particles in the water, Figure 4, giving the viewer a sort of haze or fog effect. This effect can also change the perceived color of objects reducing contrast. [2], [7], [8], [10], [11], [12]

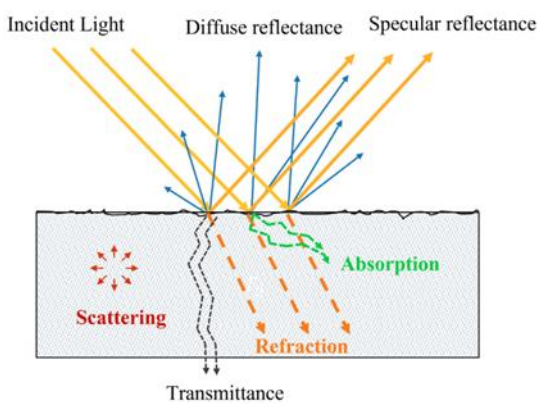

Figure 4 Schematic figure on the interaction with light with another medium. Figure from ref.[14].

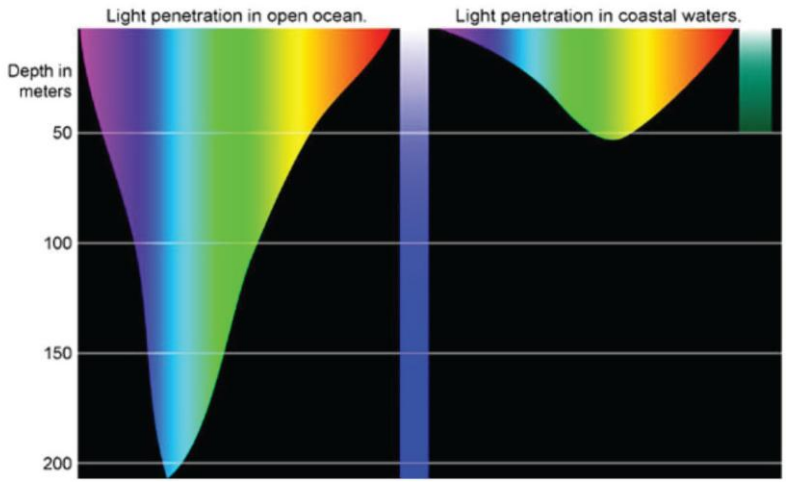

Figure 5 The depth which different colors of light penetrate ocean waters. Figure from ref. [13].

In sum, light interaction with another medium is a very complex topic, especially water, where there are several interactions at play simultaneously and where light wave lengths interact with the medium at the atomic level. For the sake of this research and topic, where the idea is to simulate underwater environments and give the user a theoretical basis to implement their artistic view, the main parameters that will be considered are absorption and scattering. Note also that all these phenomena are also modelled using several mathematical formulations, which, however, is not the goal of this literature review, and for this reason a detailed overview of the algorithms used is not given. The reader is referred to [2], [3], [4], [9], [10], [11], [12].

2. REAL TIME AND HARDWARE RENDERING TECHNIQUES

Real time and rendering techniques are two important topics to consider as well. These techniques aim at simulating underwater environments by trying to use real physical interactions to create the most realistic scene possible.[3], [11] While this is not the main focus of this topic, it gives a lot of insights on how actually these light interactions are being handled, and what parameters are more important in trying to recreate realistic underwater scenes.

For example, Iwasaki et al. (2022)[3] proposes a method to render shafts of light and caustics. They use illumination volumes for displaying shafts of light. The illumination volume is divided into several volumes so that it is suitable for graphics hardware and the intensities of the shafts of light can be calculated by using hardware color blending functions, Figure 6.[3]

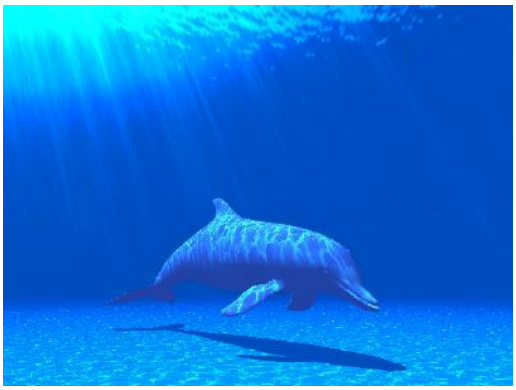

Figure 6 Example of rendered image with caustics and god rays from ref.[3].

3. ENGINE SPECIFIC IMPLEMENTATIONS

Engine specific implementations are the nuanced more artistic oriented implementation in a game engine of underwater scenes. Varying a bit from developer to developer, in techniques or tools, but always trying to address the same parameters discussed before.[4], [5], [15], [16]

In this category we see the use of directionals lights with backscatter enabled (light shafts), volumetric fog to simulate murkiness of water, shaders with textures to recreate caustics, color temperature and contrast, post processing effects to give the water a certain bluish tone, particle effects to simulate suspended particles on the water, chromatic aberration, vignettes, etc. [4], [5], [15], [16]

In figure 7 an example used in Fallout 4 of no inscattering, only scattering and combined image is given.

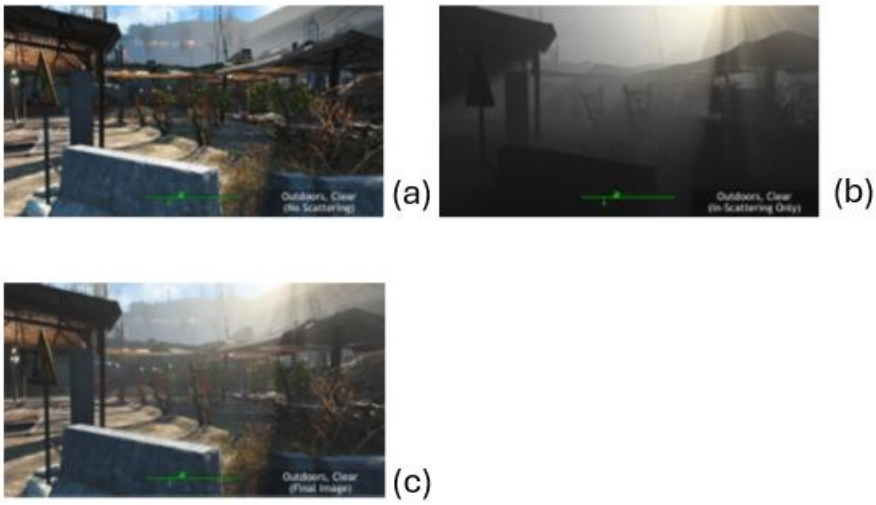

Figure 7 (a) no inscattering, (b) only scattering, (c) combined image. Images from ref. [4].

This gives an overview of what is done, in engine, to recreate the scene, and from here we can take away the main elements and parameters that are fundamental for the basis of underwater simulation. For example, we can increase the fog density, or/and use directional lights to simulate god rays. From here we can give the reader insights into what is necessary to manipulate and how to achieve their artistic goal. The next section of literature is related to this, where aesthetics is considered.

4. VISUAL/AESTHETIC CONSIDERATIONS

For visual and aesthetic purposes, playing with volumetric fog, particles and colon is key. For instance, to create a realistic underwater scene in a lake high in the mountains, then a more intense volumetric fog, particles and a more cyan color would need to be used in order to simulate the correct color, detritus and murkiness, based on the type of body of water.[8] Caustics, if used, would take into consideration the water conditions on the surface caused by weather - if calm weather then caustics would be animated more slowly to simulate the slower ripples on the surface cause by the wind.[5], [11], [17]. Depth as discussed previously, would also be represented by the attenuation of light, with a darker environment, less contrast, and more greyish tones.

If a more stylized scene is the goal, then the priority becomes exaggerating, simplifying, or abstracting reality, using distinctive shapes, proportions, textures, and colors to craft a unique visual identity.[14]

For instance, Figure 8, gives a visual example of a recreation of an underwater scene where the goal was to create a realistic scene, but where the author still took some creative liberties. We can see light shafts, blooms, particles, blue tones and even low contrasts on the corals, representing the color attenuation.

Figure 8. Underwater scene was created in Unreal Engine. Image from ref. [5].

In sum, the literature shows that in simulating underwater scenes, realism and mood are mainly composed by a combination of light absorption, scattering, and volumetric effects combined with directional lights. [5], [15], [17] Modern rendering engines recreate these physical effects into adjustable parameters such as fog density, light color temperature, caustics, post-processing effects and particle effects. [4], [5], [15], [16], [17] The insights gathered here will guide the development of a simplified framework to help artists reproduce underwater environments suited to their specific visual and emotional goals.

RESEARCH

1. METHODOLOGY

This work uses a qualitative experimental approach. By manipulating underwater scene parameters in the game engine, the aim is to see how volumetric fog, particle effects, light temperature and post processing effects influence the perceived mood and realism of the scene. This approach will then be validated by conducting a survey where several images are presented to the participants.

Unity HDRP version 6000.2.6f2 (the latest version at the time of writing) was used, that has tools such as volumetric fog, particle systems, and lightning control. The choice of using this engine was made because it is one of the most used engine by indie developers.[18]

2. HYPOTHESES

2.1. H0 – BASE SCENE

The base scene consists of a simple underwater scene with default directional light only, no volumetric fog or particles. It is expected to appear less immersive and neutral in mood and will be used as a baseline.

The baseline scene will have a terrain plain to simulate the underwater floor, set with a size o 100*100 units, at y=-50 to simulate a depth of 50 units. The terrain plain will have a simple heigh map, drawn by hand to simulate the terrain roughness and unevenness. It will also have a simple material mimicking sand for a more realistic look.

This scene will also have several mesh props, mainly rocks and coral. The rocks used will be scattered through the plane. For this I will use Rocks FREE pack by DEXSOFT from unity store. These rocks lack color variance, especially reds, so they will be mainly used for comparing visibility since they are more on the periphery of the terrain. We will be able to see how visible the rocks are when implementing our hypothesis.

For the coral, I will use "Rainbow Haven Reef - coral" (https://skfb.ly/pxGVy) by Brian Trepanier that will be on the center of the scene. This will be the main focus since its higher complexity and color is better suited for comparing attenuation and visibility.

Regarding the light, the default light from each scene will be used. It is a directional light that simulates the sun, and it will be at Position (0,100,0) with and Angle of (90,0,0). The color temperature will be 6570 (default) and the intensity will be set to maximum. Volumetrics will also be enabled.

For the camera, the scene position is irrelevant since I created a simple camera script (Fig. A 1) where I make the camera rotate around the center object (the coral). This script takes the distance, where we set the distance to the object (30), and the height, where we set the height of the camera relative to the terrain floor (15), the rotation speed (10), and the camera pitch angle (20). There is also a Boolean for the auto rotate, Figure 9.

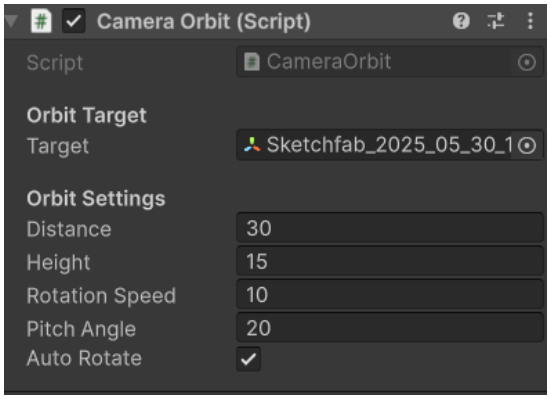

Figure 9 Camera Script parameters.

Figure 10 shows a screenshot of the baseline scene as reference with all the parameters described. This will be our baseline scene, H0, and from this all the other scenes for each hypothesis will be used.

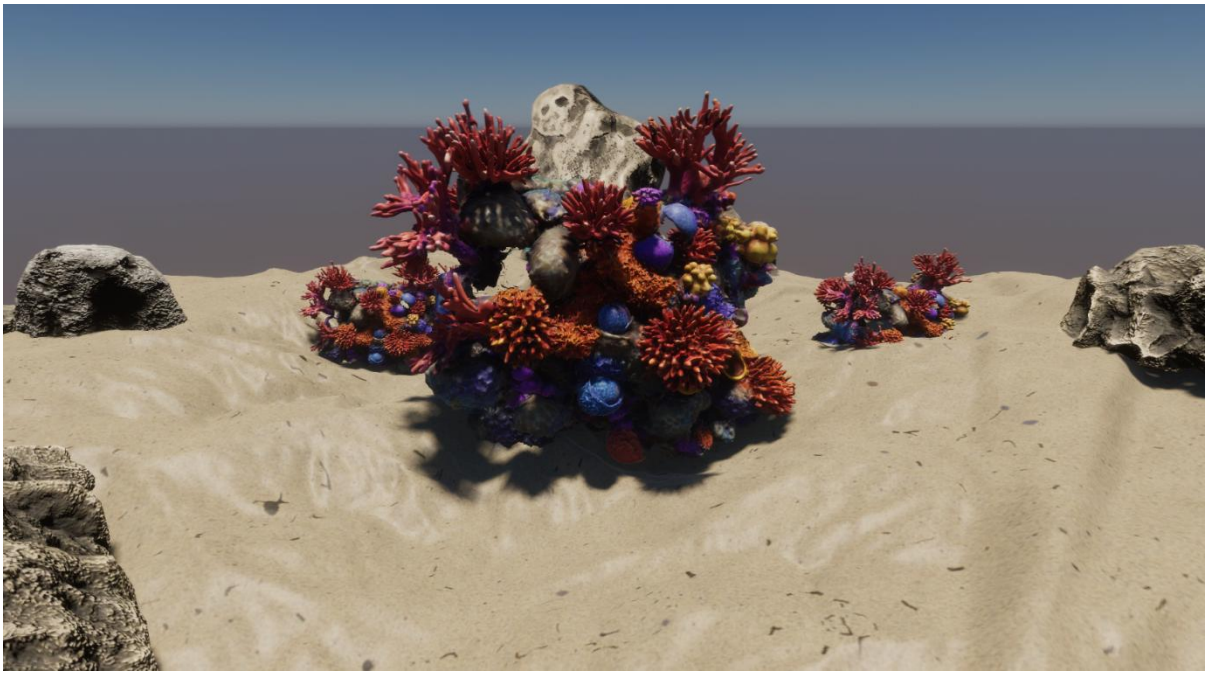

Figure 10 Screenshot of the baseline scene as reference.

2.2. H1 – ATTENUATION

In this scene a volumetric fog will be added to the scene with the main purpose of testing light attenuation. Here a Box Volume game object is added to the scene, matching the size of the terrain and depth. This will be set as a local volume, with added fog override, Figure 11.

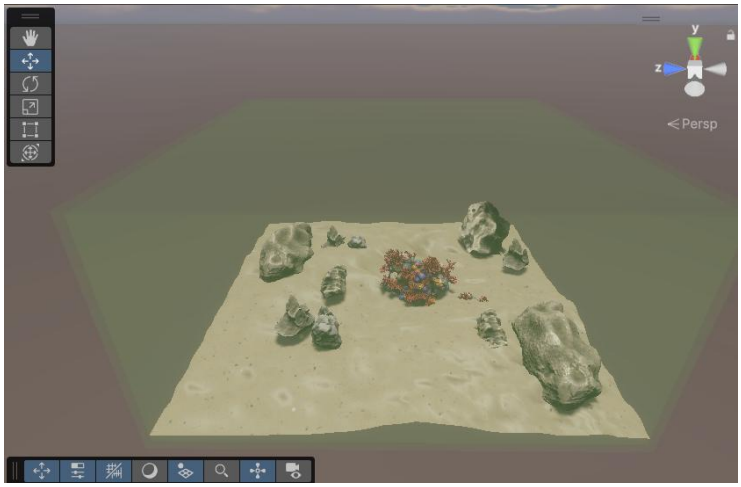

Figure 11 Volume box size in scene.

The fog is set to enabled, enabling the parameters as seen in Figure 12.

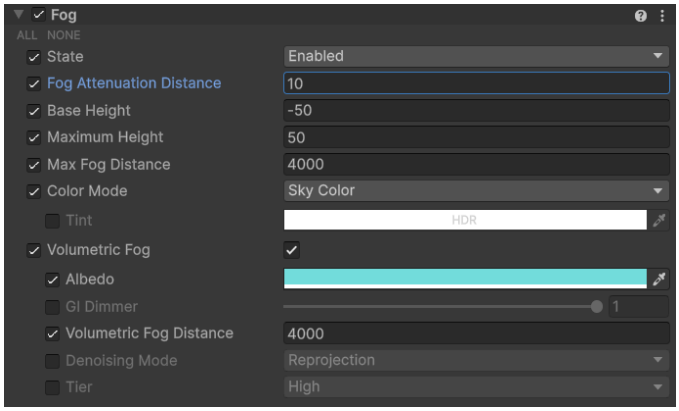

Figure 12 Used Fog parameters.

Here Volumetric fog must be enabled, and the main parameter to analyze will be the fog attenuation, which controls the density at base level the Distance at which fog reduces background light intensity by 63%. So, when we reduce the Fog Attenuation Distance, we increase its density, increasing light attenuation, mainly scattering. I also set an albedo color to a light blue/green color to simulate absorption. This gives a blue hue to the fog simulating underwater visibility.

Figure 13 With fog attenuation 10. (less visibility/more light attenuation).

Figure 14 With fog Attenuation 25. (higher visibility/lesser light attenuation).

2.3. H2 – HAZE/MURKINESS

In this scene we will simulate further the scattering of the volumetric fog to simulate real scattering of the light rays in the water, to create a haze effect to reduce visibility and change perceived mood. We will also add particles

effect to the water, simulating suspended detritus in order to increase the scene realism. For this, Bloom will be added to the volume override with the parameters enabled as seen in Figure 15.

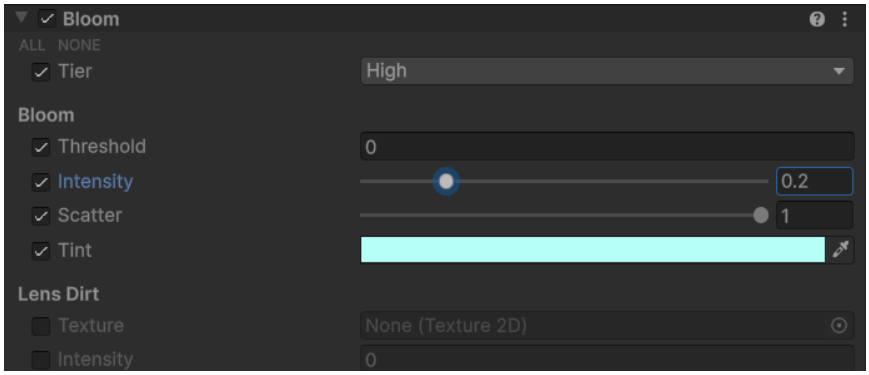

Figure 15 Enabled parameters for Bloom.

Here we increase the scattering effect of the fog, decreasing visibility, giving it a more murky/ hazy environment. A tint can also be applied to further enhance the color absorption, Figure 16 and 17.

Figure 16 Bloom intensity 0.2 (less hazy/murky).

Figure 17 Bloom intensity 0.8 (more hazy/murky effect).

For the particle effect [8], this will add realism to the scene, where we can simulate suspended matter. This also contributes to giving the feeling of movement to the water. The screenshot (Figure 18) shows the parameters used for the particle effect and Figure 19 shows the number of particles in scene.

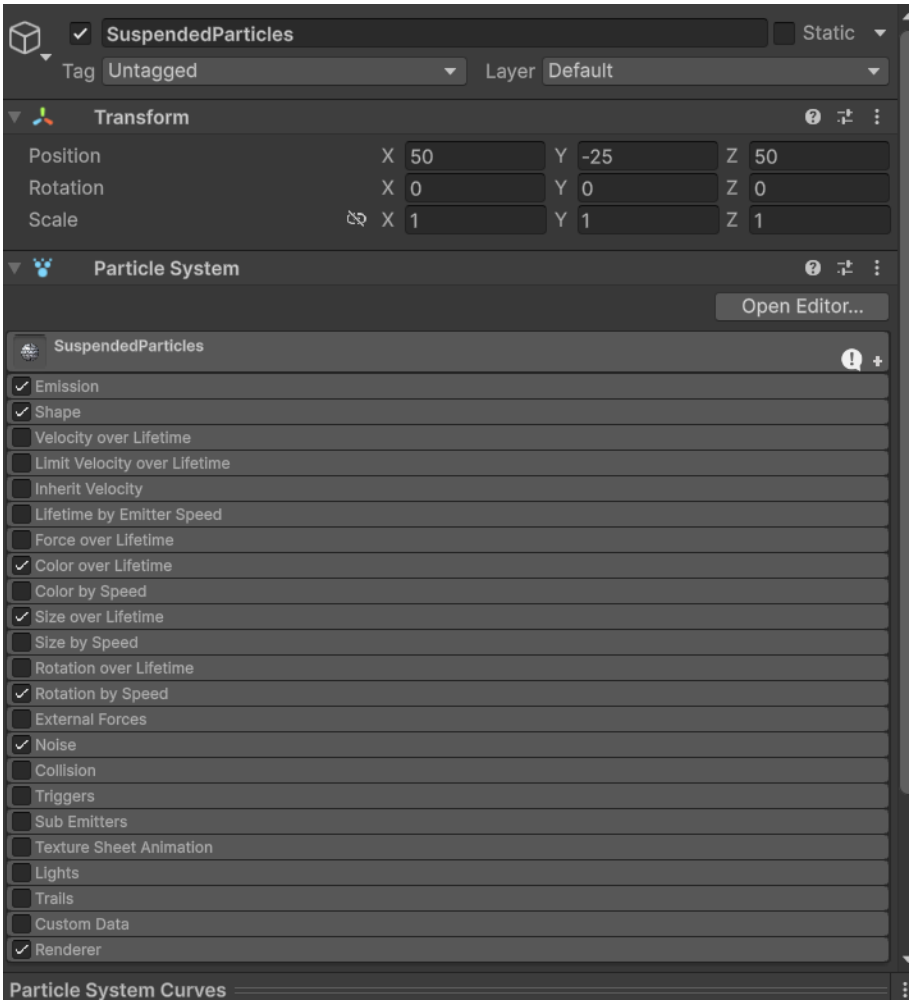

Figure 18 Used parameters for the particle effect.

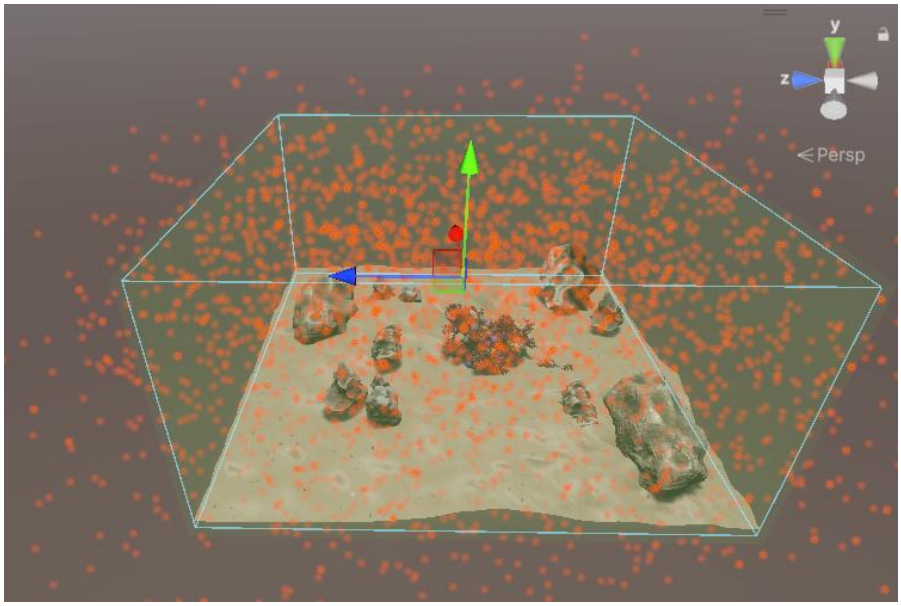

Figure 19 Visualization of number of particles in Scene.

We set its shape as a box volume so that it fills the scene area, and we also use the default material for the particle system, in a sphere mesh. We set their speed to 0.1, so it simulates suspended particles, and adds noise for some random movement. We can increase the number of particles by changing their emission, Figure 20 and 21.

Figure 20. With emission 200 (higher particles, 20000 in scene).

Figure 21. With emission 100 (higher particles, 10000 in scene).

2.4. H3 – COLOR TEMPERATURE

In this scene the directional light temperature and intensity that simulates the sun to change the mood of the scene will be adjusted. We expect to see a change in perceived mood. If warmer tones are used, we can expect a more relaxed tropical mood, if colder tones are used, we can expect a more mysterious mood. Some post processing effects can be added to the Volume of the fog like Color Adjustments and Split Toning to further enhance and change the mood of the scene, Figure 22 and 23.

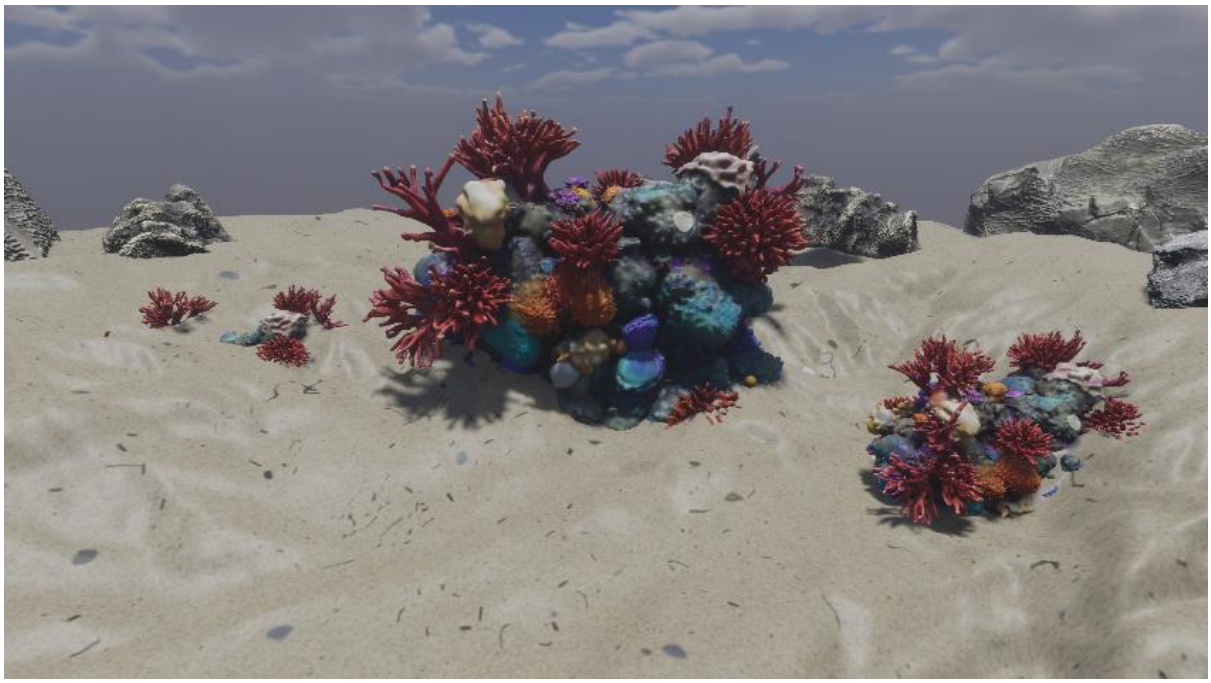

Figure 22. With 9000K temperature.

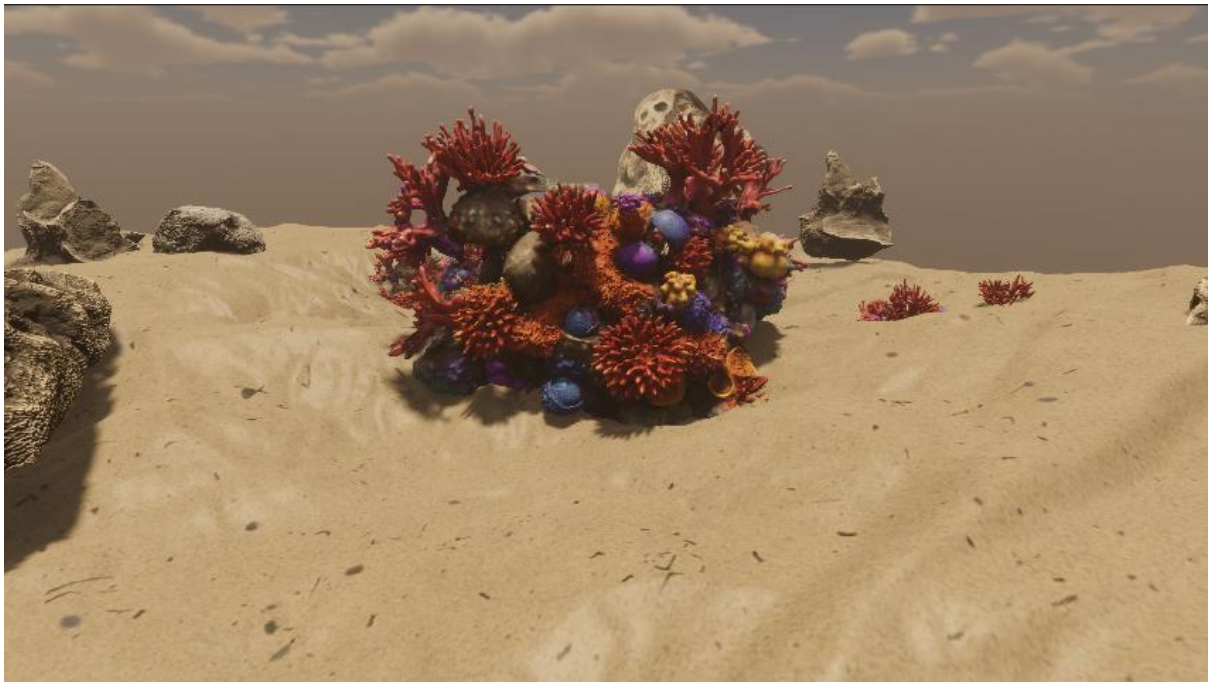

Figure 23. With 5000K temperature.

With Color Adjustment we can decrease contrast and saturation further, simulating light absorption. This would benefit from a script, where the contrast would decrease based on depth. By using Tone Mapping, different areas of the image are tinted based on luminance values, to help achieve a more distinctive look.

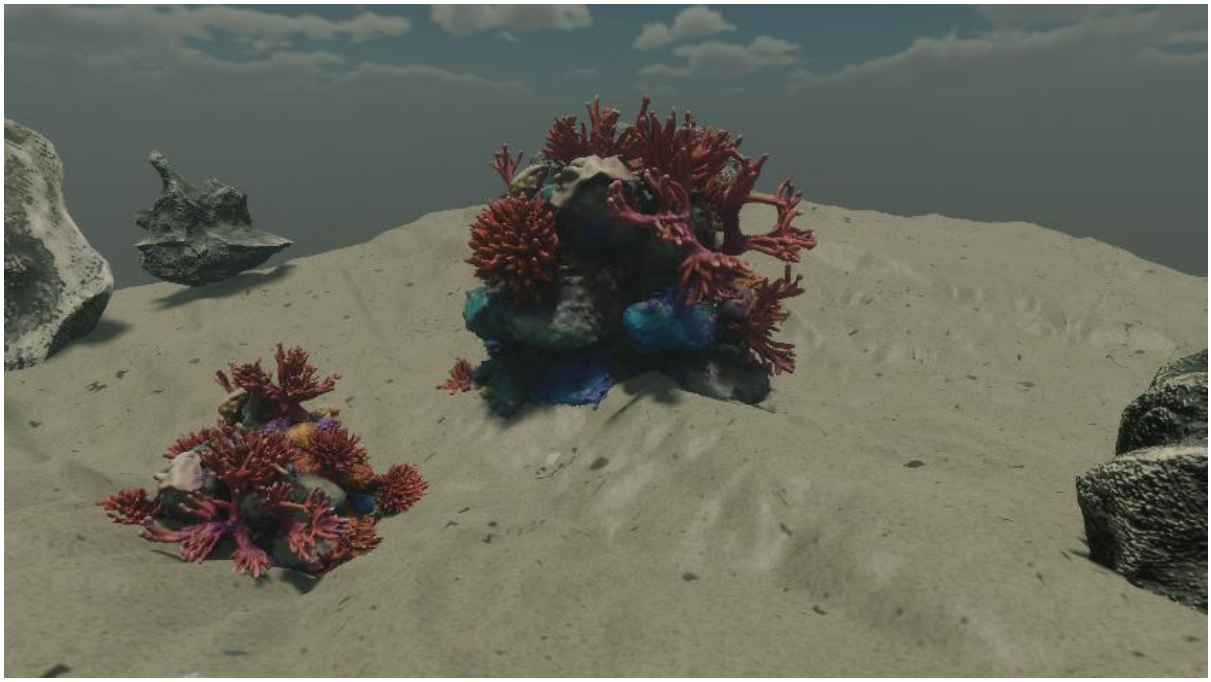

Figure 24. Light Temperature set to 5000K with Color Adjustment and tone mapping enabled.

Figure 25. Light Temperature set to 8000K with Color Adjustment and two-tone mapping enabled.

2.5. COMBINED SCENE

This combined scene is the final step in our analysis, and I will use a real-life image of 2 Optical Water Types (OWT)[8] (Table A 1): OWT 2 (typically found in open ocean tropical or subtropical regions) and OWT 4a (usually found in coastal zones and inland water bodies). This will be the reference for our combined scenes.

2.5.1. REFERENCE IMAGES FOR COMBINED SCENES

Figure 26. OWT 4a reference image.

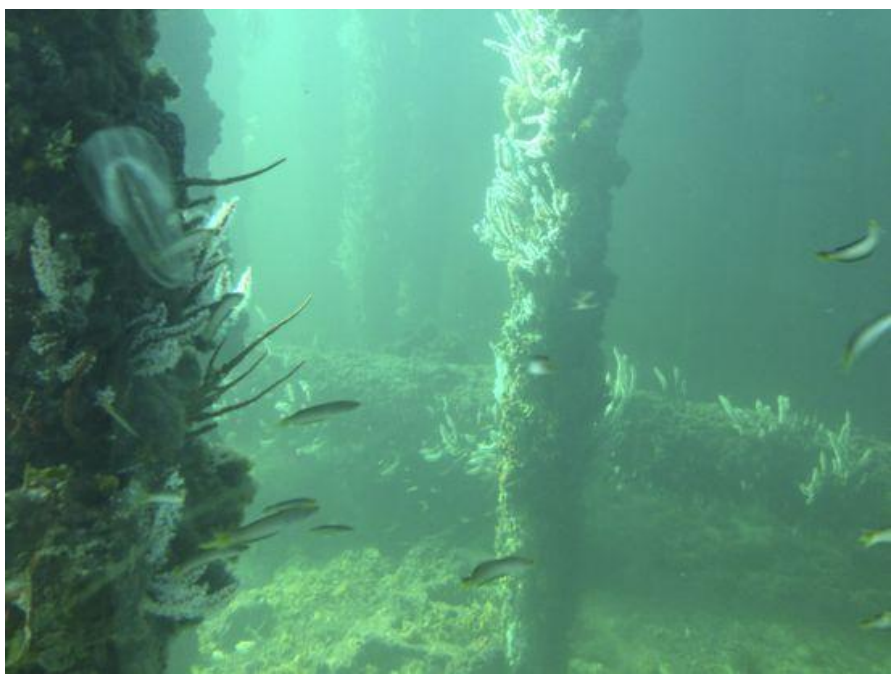

Figure 27. OWT 4a reference image.

Figure 28. OWT 4a reference image.

Figure 29. OWT 2 reference image.

Figure 30. OWT 2 reference image.

Figure 31. OWT 2 reference image.

This scene will be the result of the combination of all previously stated hypotheses where we will get a combined result that should be visually cohesive scene with realistic attenuation, subtle murkiness, and mood appropriate color temperature representing the balance between realism and artistic goals.

2.5.2. COMBINED SCENE – OWT 2

In this scene we try to replicate based on the reference images (Figure 29 to 31) an underwater scene based on the OWT 2 (typically found in open ocean tropical or subtropical regions). For this scene we want to have higher visibility bluish tones and use warmer light for a calmer mood.

For the H1 part of this scene we will use a higher light attenuation distance, which implies less light attenuation and a blue color like the reference scene, Fig. 32.

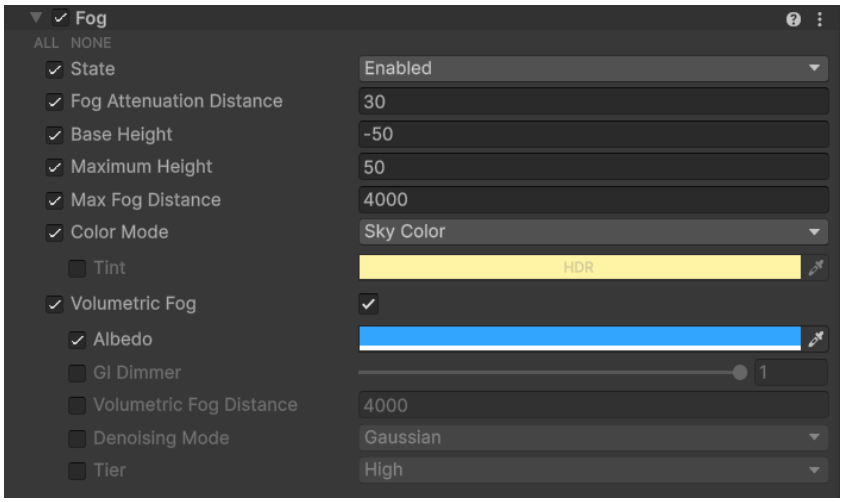

Figure 32. Combined scene H1 (OWT 2) parameters (volumetric fog).

For the H2 part of this scene (Bloom and particle effects) I will use a weaker bloom effect and less particles (Fig. 34), so it represents more clear water and less detritus. The bloom intensity is set to 0.5 but with a lower scatter so the image is more nitid, Fig 33.

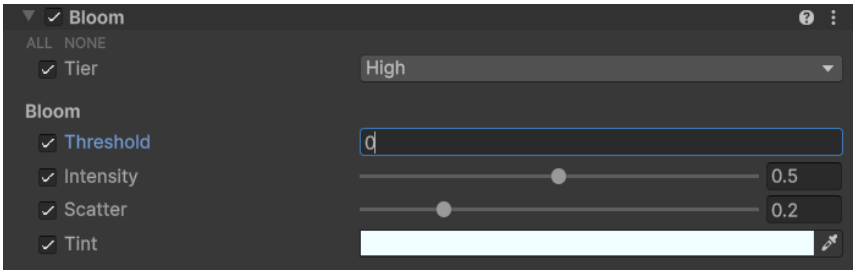

Figure 33. Bloom parameters.

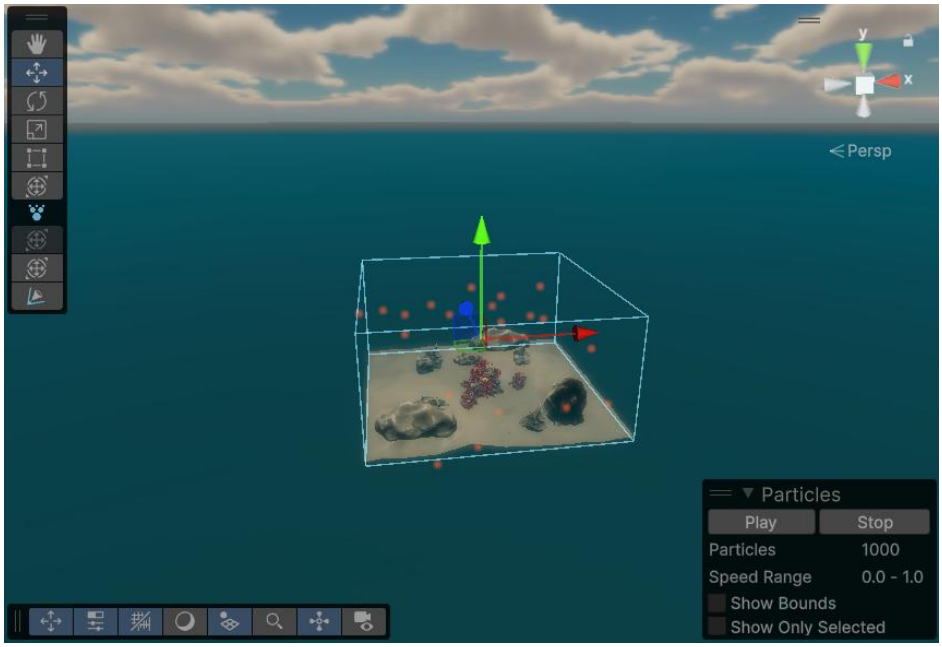

Figure 34 Particles in scene.

For the H3 (color temperature and post processing effect) part of this scene we set the light temperature to 5000K to represent warmer tones in a tropical environment and use Color Adjustments to increase contrast and saturation to make the color more intense (Fig. 35), simulating the lower depths of the reference images where there is not much light absorption.

Figure 35. Color Adjustment and Split Toning parameters.

After combining these parameters, we get out combined scene, which aims to represent the reference images of OWT 2, a tropical underwater scene with high visibility and warmer tones, invoking a calmer mood, Fig. 36. This is the image from the scene that will be used for the survey where we can gather the data and answer our research questions.

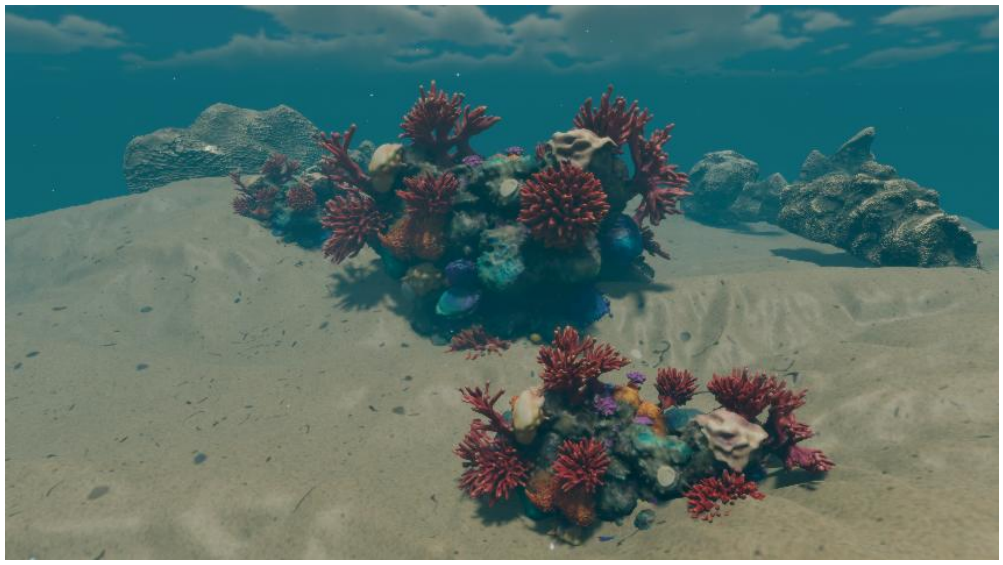

Figure 36 Combined scene for OWT 2.

2.5.3. COMBINED SCENE – OWT 4A

Here we do the same process as before for this scene. where we try to replicate based on the reference images (Figure 26 to 28) an underwater scene based on the OWT 4a (usually found in coastal zones and inland water bodies). This scene we want to have lower visibility, greenish tones and use colder light tones for a more Tense mood.

For the H1, we use a lower fog attenuation distance (higher light attenuation) to simulate lower visibility of the scene, and a green color to the fog, matching OTW 4a, Fig. 37.

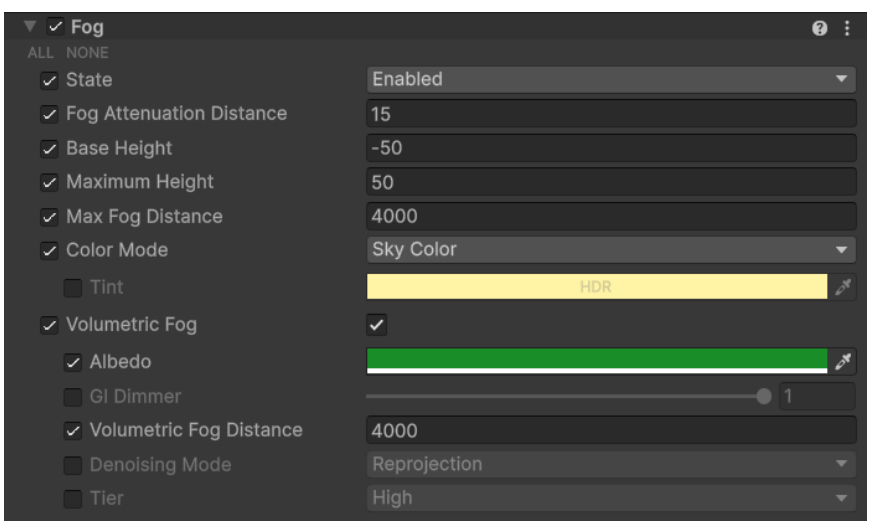

Figure 37 Combined scene H1 (OWT 4a) parameters (volumetric fog).

For the H2 part of this scene (Bloom and particle effects) I will use a stronger bloom effect and more particles (Fig. 39), so it represents less clear water (less visibility) and more detritus. The bloom intensity is set to 0.8 with a higher scatter compared to the combine scene 1 so the image is blurrier, Fig 38.

Figure 38 Bloom parameters.

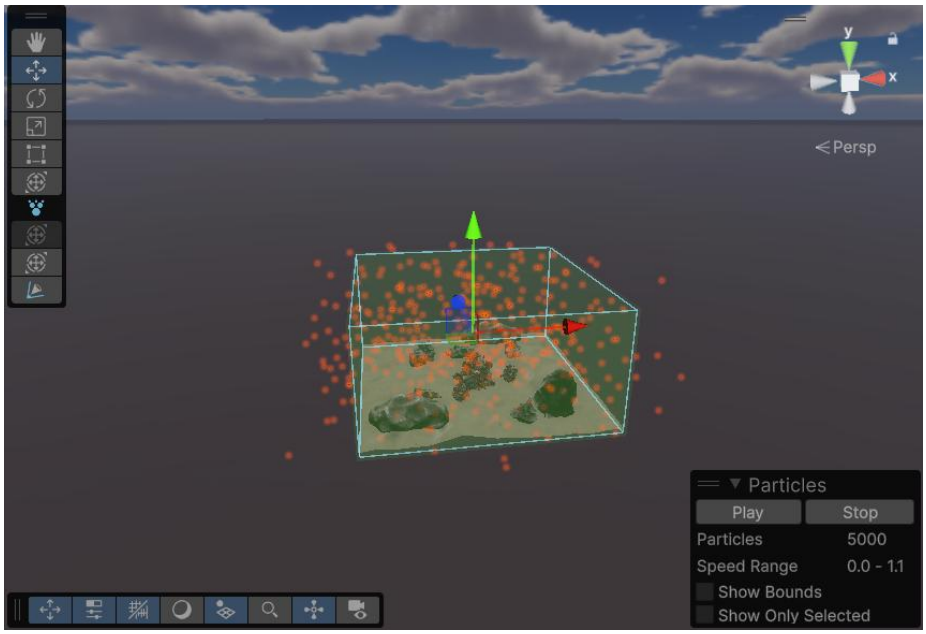

Figure 39 Particles in scene.

For the H3 (color temperature and post processing effect) part of this scene we set the light temperature to 8000K to represent colder tones in a coastal environment and use Color Adjustments to decrease contrast and saturation to make the color less intense (Fig. 40), simulating the higher depths of the reference images where there is more light attenuation.

Figure 40 Color Adjustment and Split Toning parameters.

After combining these parameters, we get out combined scene, which aims to represent the reference images of OWT 4a, a coastal or river underwater scene with lower visibility and colder tones, invoking a more tense mood, Fig. 41.

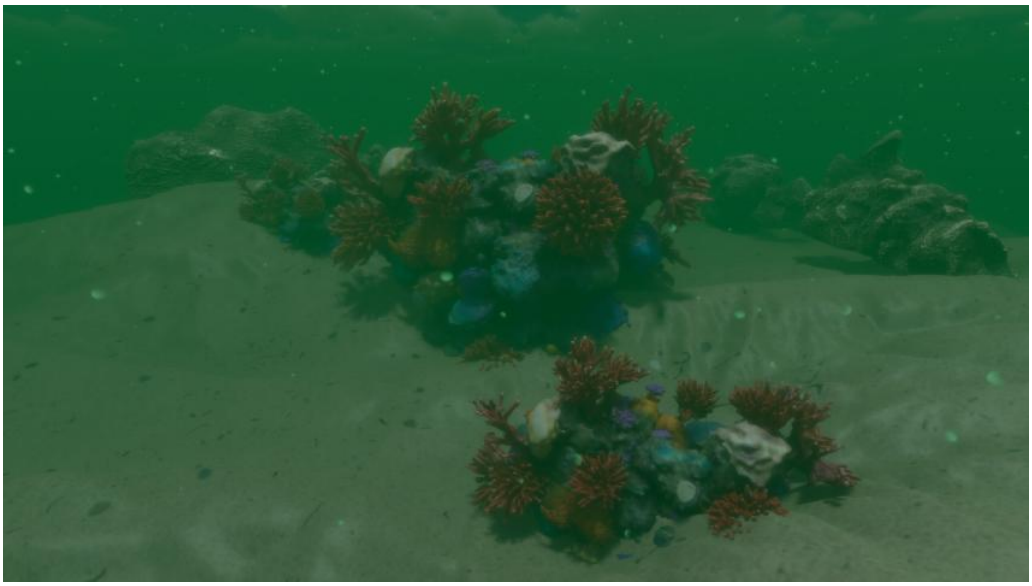

Figure 41 Combined scene for OWT 4a.

3. DATA COLLECTION METHOD

The data collection method was done by survey. A minimum of 25 participants were asked to answer a few questions regarding the hypothesis scenes and the combined scenes.

The survey was split into several sections, each section related to an hypothesis and finally the combined scene.

Each section had several questions that are relevant to the hypotheses in question. H1 and H2 asked about the scene visibility, H3 about the scene mood, and the combined scene all the questions asked previously.

Once the data from the survey was collected, the research question and the hypotheses "*How does volumetric fog affect perceived depth and realism? How does particle density and distribution influence visibility and immersion? How does directional light temperature modify emotional perception?"* could be answered from a user's viewpoint. The data survey gives valuable input from an external user perspective by incorporating the subjectivity aspect in these visual effects. Thus, the data collected from the survey can be used to get insights on the subjectivity of visuals and to improve the simulation.

3.1. SURVEY PART 1 – H1

In this section, the participants were asked to answer about the scene's visibility for two different images: "How clear is the underwater environment in terms of visibility?" One of the images is related with high attenuation distance of the volumetric fog (higher visibility, Fig. 14) and the image is related to low attenuation distance (lower visibility, Fig. 13). The participant can choose an answer between 1 and 5, where 1 is low visibility, and 5 is high visibility.

3.2. SURVEY PART 2 – H2

In this section, the question is the same as in the previous section, but the images shown are related to the hypothesis H2 of bloom effect + particle count. One of the images is related with weak bloom effect + low particle count (Fig. 16) and the other image is related with strong bloom effect + high particle count (Fig. 17). The participant can choose an answer between 1 and 5, where 1 is low visibility, and 5 is high visibility.

3.3. SURVEY PART 3 – H3

In this section, the participant is questioned about the mood of the scene, with the question being: What mood does this evoke? (Calm to Tense).

Two images are presented, one is related with Lower Color Temperature (Colder, Fig. 22) and the other is related with Higher Color Temperature (Warmer, Fig. 23). The participant can choose an answer between 1 and 5, where 1 is Calm, and 5 being Tense.

3.4. SURVEY PART 4 – COMBINED SCENE

This part is split into two, with each part related to a scene corresponding to the respective OWT type: Fig 36 for the OWT 2 scene, and Fig. 41 for the OWT 4a scene. The survey uses real life images of OWT 2 and OWT 4a as reference (Fig. 26 to 31) and then each scene has the same four questions. For each question the participant will be shown an image that is the result of the combination of the hypotheses for each scene. This part of the survey is intended to check the aggregated result of all the hypotheses together and their impact on the viewer. For

example, will the change in Color temperature of light have the expected effect on the user in the combined scene the same way that in H3.

3.4.1. WHICH REAL-WORLD WATER TYPE DOES THIS SCENE RESEMBLE?

Here the participant can compare the provided images and choose between OWT 2 and OWT 4a. This question is "graded", so that it is possible the analyze if the participant perceives and matches the image given to their real-life counterpart.

3.4.2. HOW CLEAR IS THE UNDERWATER ENVIRONMENT IN TERMS OF VISIBILITY?

The question will be the same as the hypotheses for H1 and H2 in part 1 and 2, respectively.

3.4.3. WHAT MOOD DOES THIS EVOKE? (CALM TO TENSE)

The question will be the same as the hypothesis H3 in part 3.

3.4.4. HOW REALISTIC DOES THIS SCENE LOOK?

Here the participant can answer from 1 – Not realistic to 2 – Very realistic.

RESULTS AND DISCUSSION

The survey was conducted for two days using google forms, and the total number of participants was 33. The median was used as the primary measure of central tendency, while the mean was included to indicate overall trends using excel from the exported data of the survey.

1. H1 – ATTENUATION

In the first section of survey the results show a median answer of 4 for the image with higher fog attenuation distance (higher light attenuation) and a median answer of 2 for the image with lower fog attenuation (higher light attenuation) where 1 is less visibility and 5 higher visibility, Figure 42 and 43. The results validate Hypothesis 1 where volumetric fog attenuation strongly affects perceived underwater visibility.

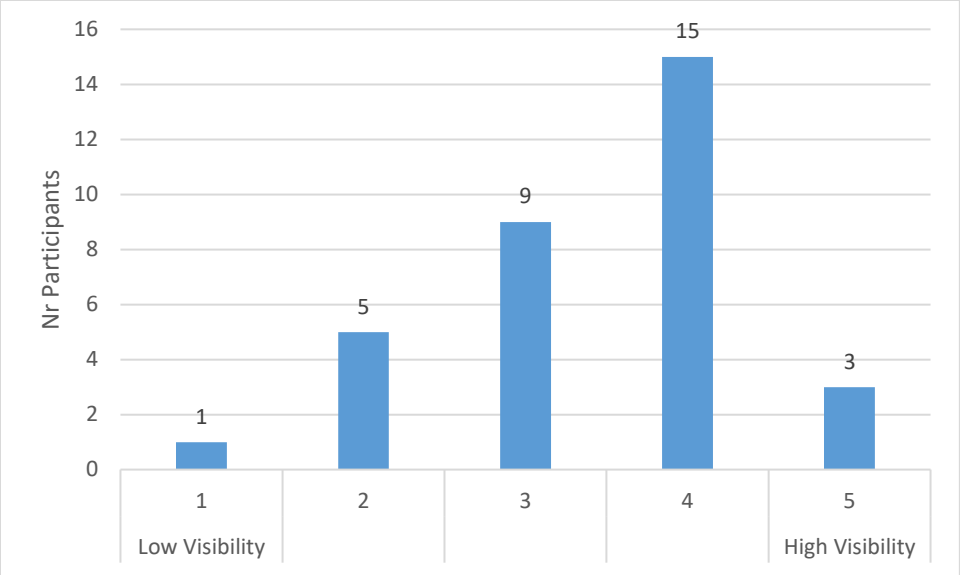

Figure 42 Question: "How clear is the underwater environment in terms of visibility" – related with lower light attenuation.

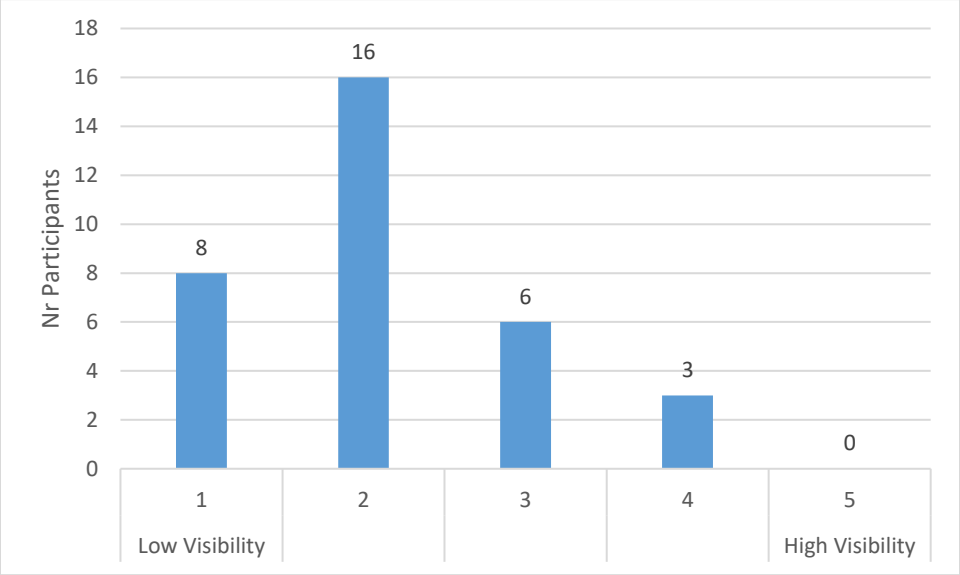

Figure 43 Question: "How clear is the underwater environment in terms of visibility" – related with higher light attenuation.

2. H2 – HAZE/MURKINESS

In the second section of the survey the results show a median answer of 2 for the image with higher Bloom effect (lower visibility) and a median answer of 5 for the image with lower Bloom effect (higher visibility) where 1 is less visibility and 5 higher visibilities, Figure 44 and 45. These results validate the Hypotheses 2 where Bloom and particles reduce clarity for most survey participants.

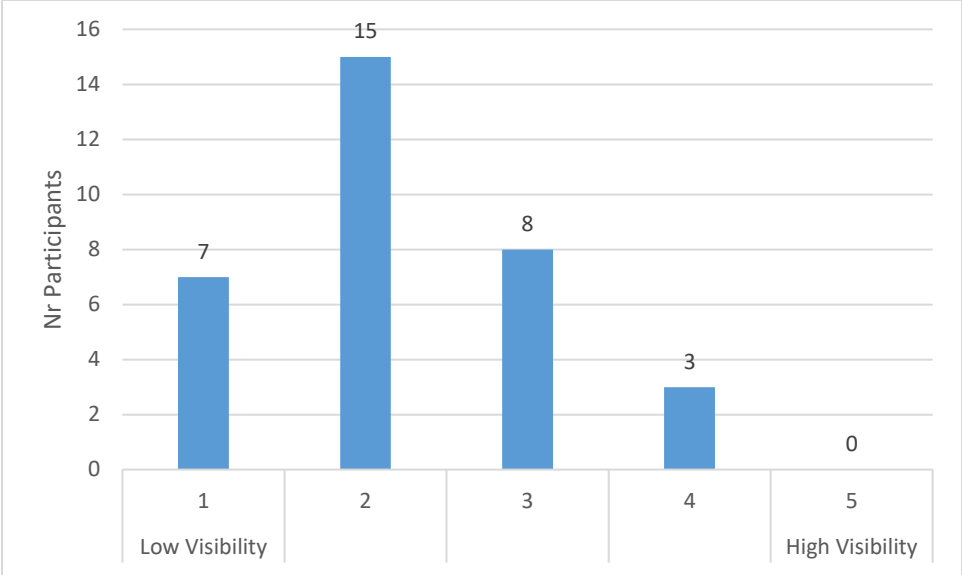

Figure 44 Question: "How clear is the underwater environment in terms of visibility" – related with stronger bloom effect.

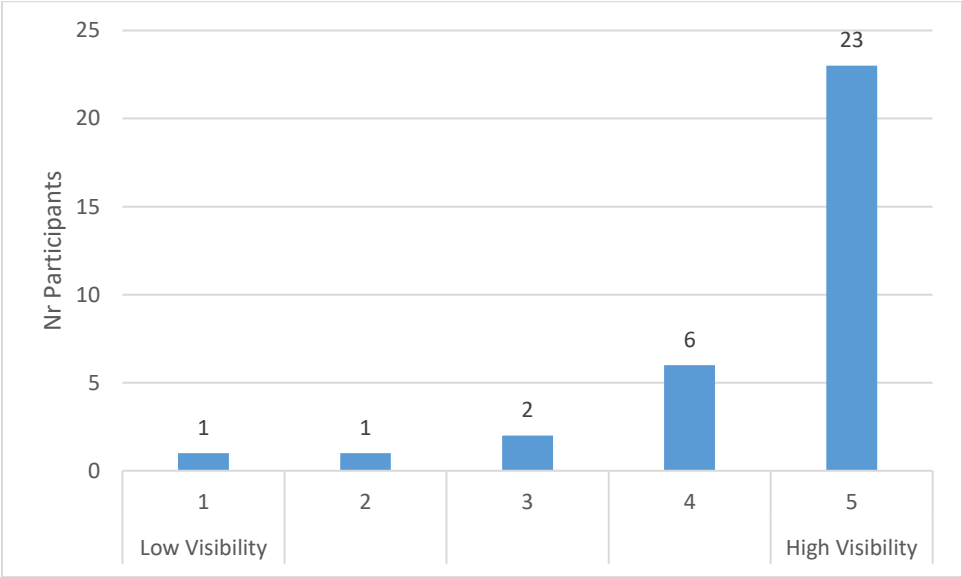

Figure 45 Question: "How clear is the underwater environment in terms of visibility" – weaker bloom effect.

3. H3 – COLOR TEMPERATURE

In the third section of the survey the results show a median answer of 3 for both images with warmer (5000K) and colder (8000K) light color temperatures. However, if we compare the mean, then it is possible to see some tendency that points in the H3 direction. For the image with warmer light, we have a mean of 2.9 which slightly points in the direction of a calmer mood. For the image with colder light temperature, we have a mean of 3.2 which also slightly points in the direction of a more tense scene, Figure 46 and 47.

While the median value is identical in both questions, median 3 (neutral value), indicates that, for most participants, changes in color temperature alone did not lead to a strong shift in emotional perception, the mean might give small tendency insight for a weak relationship between color temperature and scene mood.

These results imply that color temperature may influence emotional perception to a limited extent, but that its effect is not strong when considered in isolation.

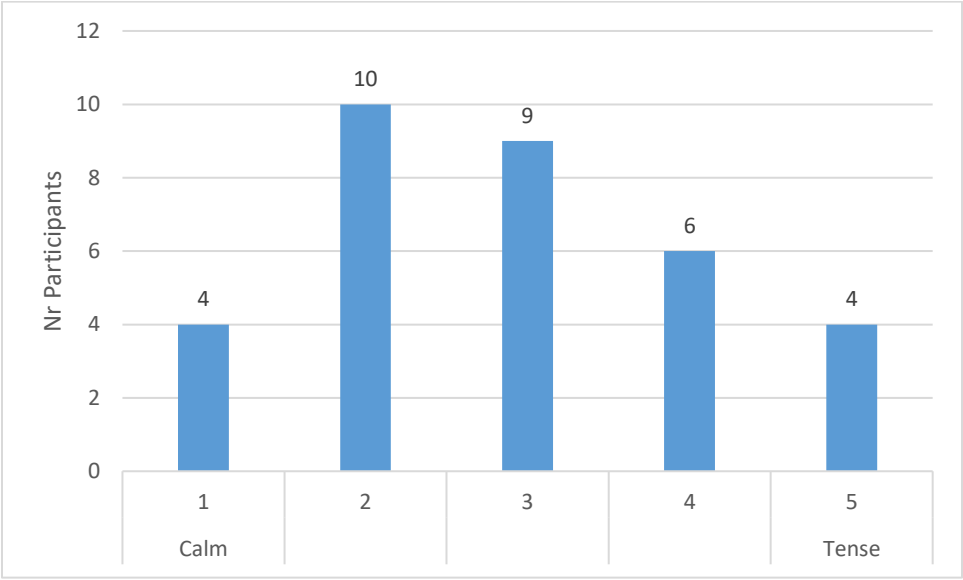

Figure 46 Question: "What mood does this evoke? (Calm to Tense) – related with warmer color temperature.

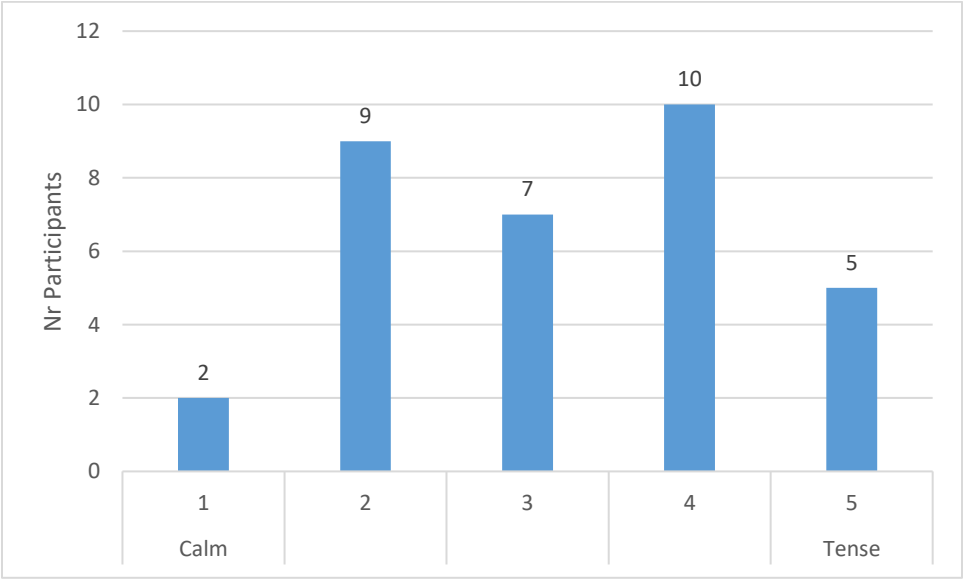

Figure 47 Question: "What mood does this evoke? (Calm to Tense) – related with colder color temperature.

4. COMBINED SCENES

In the last section of the survey, for the first question, *Which real-world water type does this scene resemble?* only 42% of participants chose the correct choice (OWT 2, Fig. 48). This result was unexpected, especially if you compare it to the results of the same question for Scene 2, where 85% of participants chose correctly (OWT 4a, Fig. 49). I think that without further feedback from the participants it is hard to give a definitive answer, but still, there

are some conclusions we can draw from this result. When analyzing the results, I reached out to some of them in order to understand their reasoning behind their choice and asked them to expand on their reasoning, and the answers obtained relate to the fact that they only felt that the image from OWT 2 scene where more similar to the reference images of OWT 2 after they answered the same question regarding the OWT 4a scene. Some even suggest that if the question for the OWT 4a scene was first and not second, they would not have made that choice.

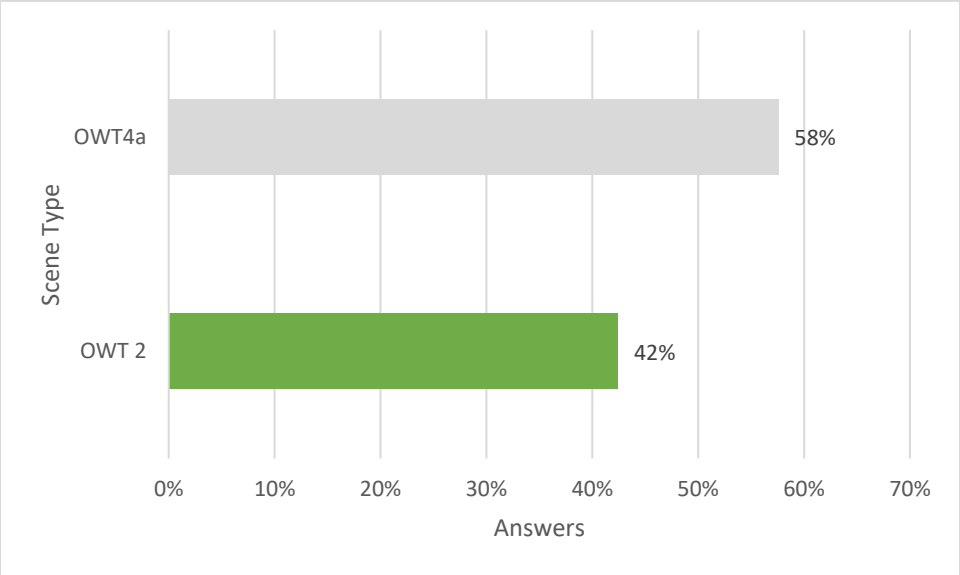

Figure 48 Question: "Which real-world water type does this scene resemble"- related with OWT 2 scene.

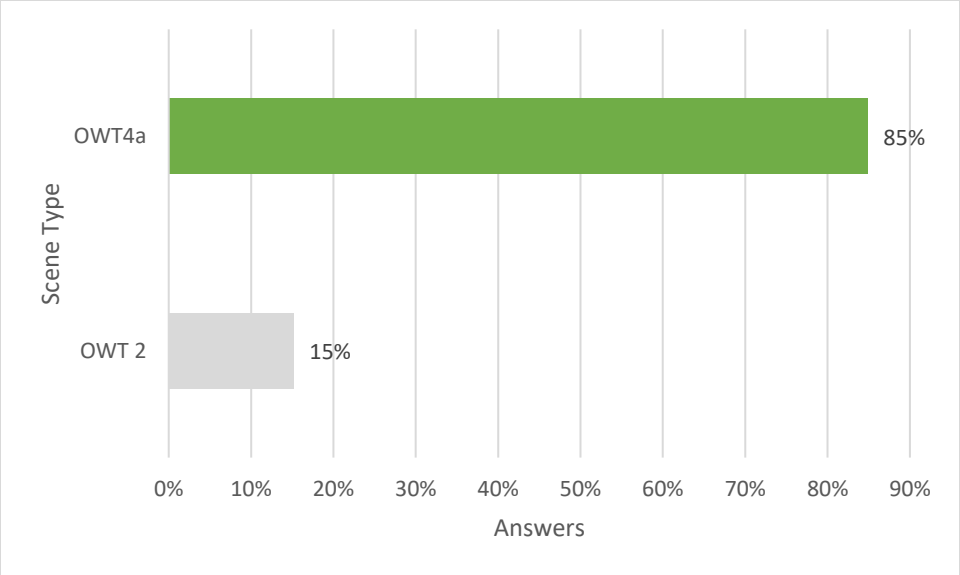

Figure 49 Question: "Which real-world water type does this scene resemble"- related with OWT 4a scene.

These results may indicate that the survey had limitations, and the way the images were presented could have been different. Also to be considered is that simply the OWT 2 scene was not an accurate depiction of the OWT 2 real life image. This can be related with the fact that the assets used and scene setup, or even the reference images chosen, were not correct, and that the parameters that were tested and used, are not particularly relevant when trying to recreate a OWT 2 underwater scene.

For the question: *How clear is the underwater environment in terms of visibility?*, Figure 50 and 51, it is possible to see that for the OWT 2 scene the median answer was 4, and for the OWT 4a it was 2. This clearly shows that in the combined scene the use of volumetric fog attenuation distance bloom effect, have a clear effect on the perceived visibility of the participant.

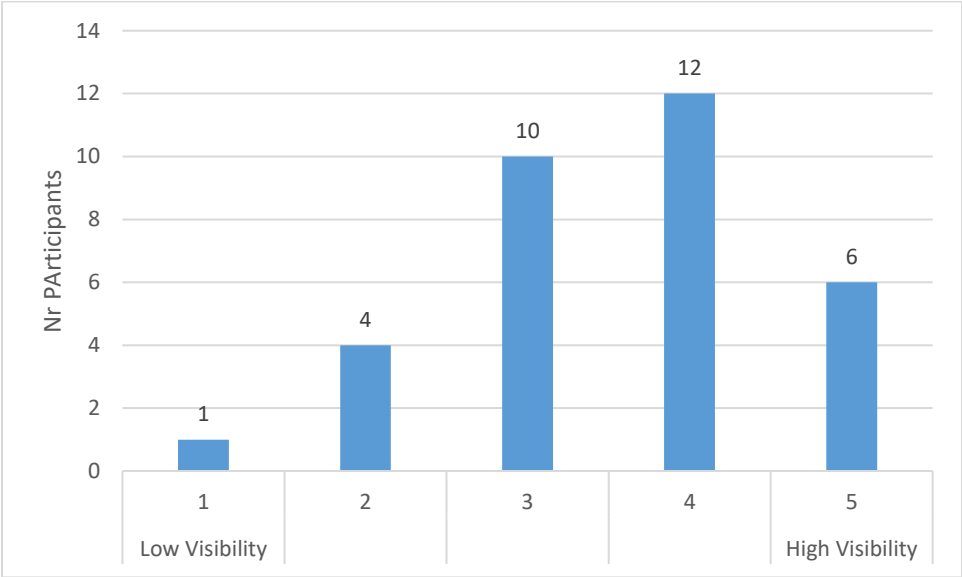

Figure 50. Question: "How clear is the underwater environment in terms of visibility" – related with OWT 2 visibility.

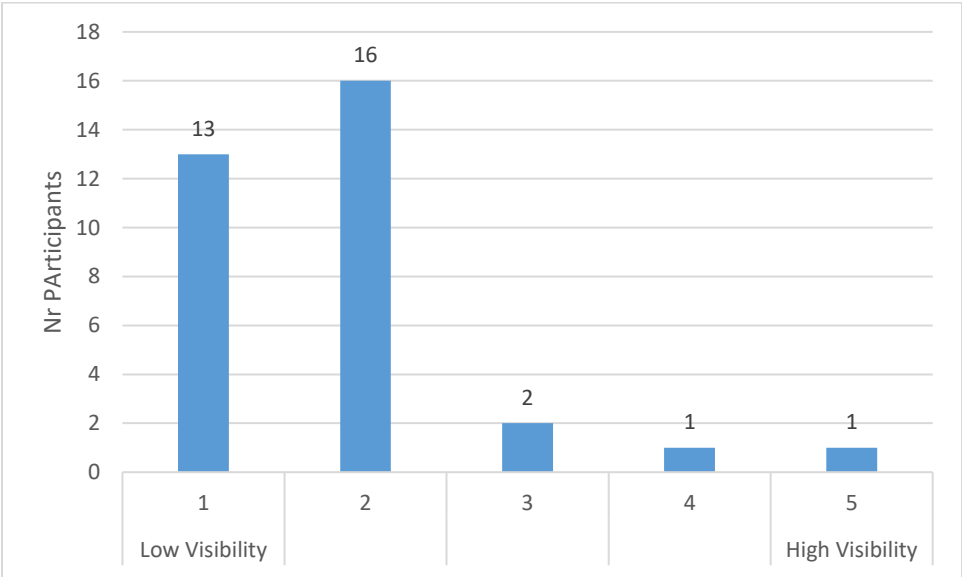

Figure 51. Question: "How clear is the underwater environment in terms of visibility" – related with OWT 4a visibility.

For the question: *What mood does this evoke? (Calm to Tense)*, Figure 52 and 53, the results show that for the OWT 2 scene, the median value was 3 with a mean of 2.7, which shows an tendency to a calmer mood. For the OWT 4a, the median answer was 4 with a mean of 3.5, which shows a tendency to a more tense mood.

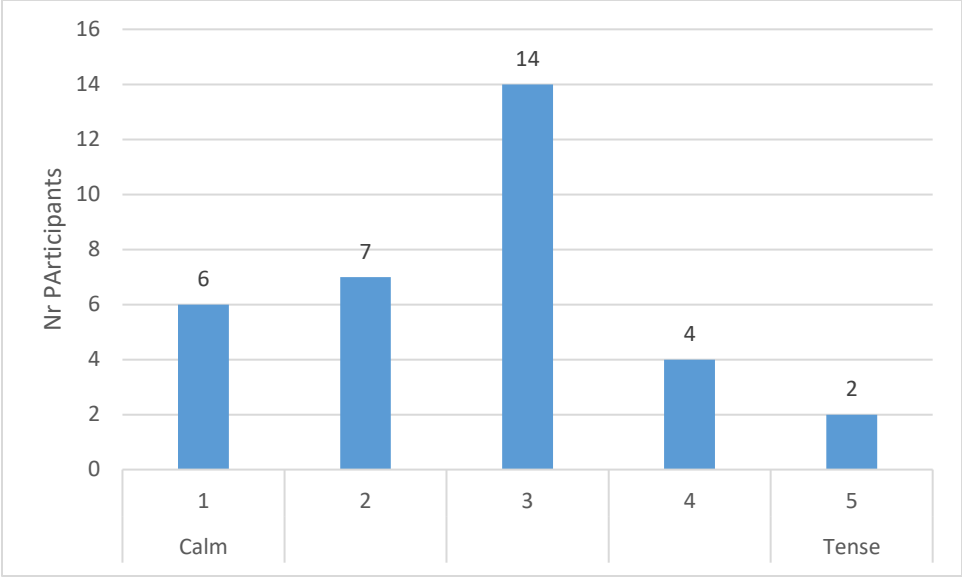

Figure 52. Question: "What mood does this evoke? (Calm to Tense) – related with OWT 2 scene mood perception.

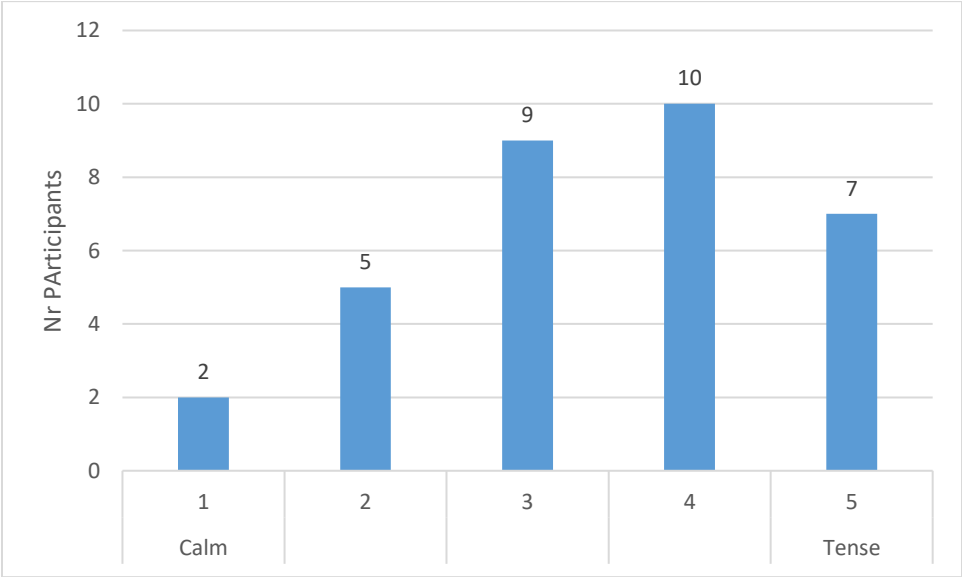

Figure 53. Question: "What mood does this evoke? (Calm to Tense) – related OWT 4a scene mood perception.

This shows that in the combined scene the answers related to the mood are the same as the isolated H3 answer, but with a stronger tendency in the OWT 4a scene to a more tense scene. This way we can conclude that visibility also has a strong effect in the participants' mood perception.

Regarding the question: *How realistic does this scene look?*, Figure 54 and 55, OWT 2 and OWT 4a scene have a median answer of 3 (1 not realistic to 5 very realistic). If we look at the mean value for tendency, we see that the OWT 2 scene has a mean answer of 3.3, while the OWT 4a has a mean answer of 2.8. We can see that there is an impact on the perception of realism when manipulating these parameters.

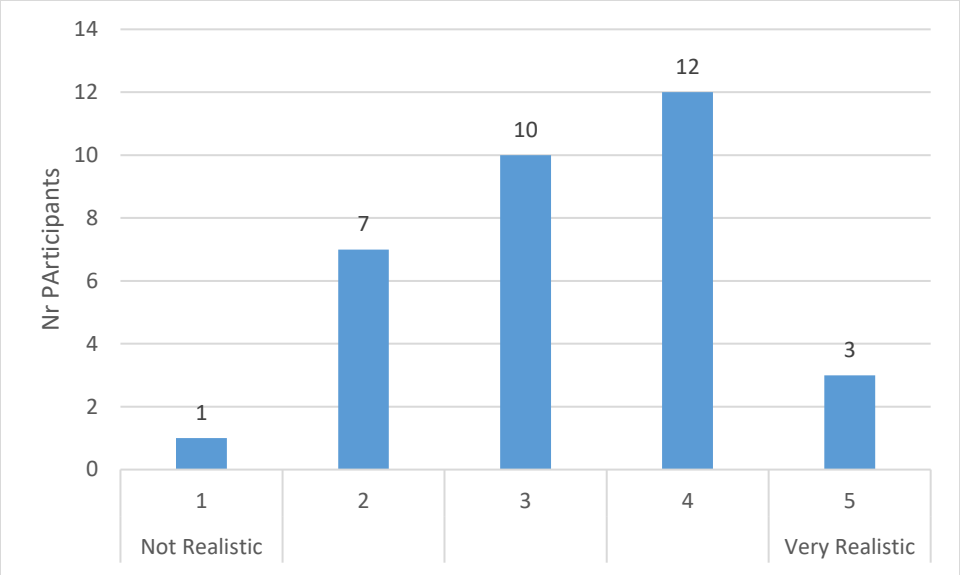

Figure 54. Question: "How realistic does this scene look?" – related with OWT 2 scene realism perception.

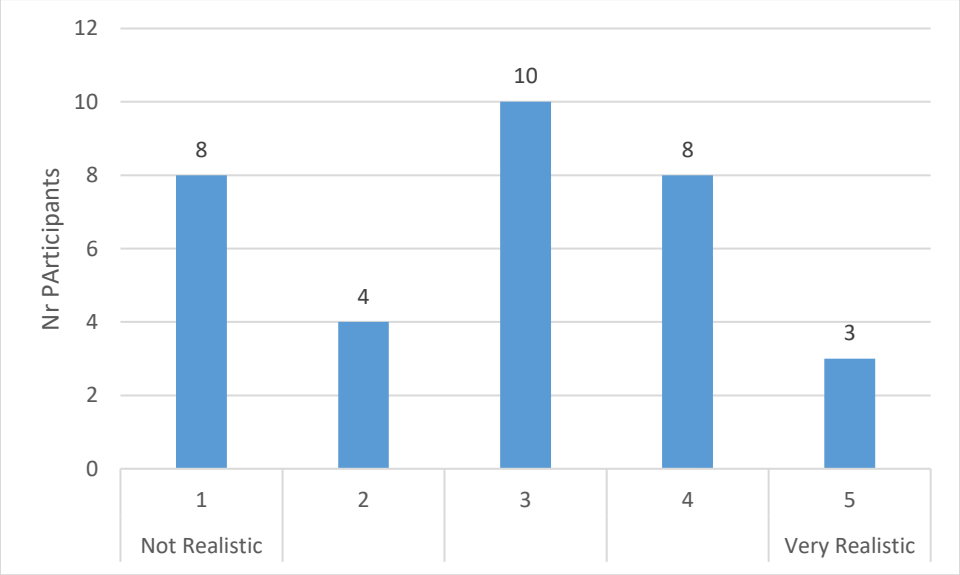

Figure 55. Question: "How realistic does this scene look?" – related OWT 4a scene realism perception.

CONCLUSION

This study investigated how volumetric fog attenuation, bloom and particle density, and directional light color temperature and color adjustments influence perceived visibility, mood, realism, and real-world water type recognition in underwater scenes. Using a two-day survey with 33 participants, perceptual responses were analyzed primarily through median values, with mean values used to identify overall tendencies.

As stated at the start of this work, the main research question is: *how do specific lighting and volumetric parameters (fog density, color absorption, particle density, and light temperature) influence the perceived mood and plausibility of underwater scenes?* From this the question was split into simple ones for our hypotheses: *How does volumetric fog affect perceived depth and realism? How does bloom and particle density and distribution influence visibility and immersion? How does directional light temperature and color adjustments modify emotional perception?*

The survey results show that volumetric fog attenuation and bloom effects have a strong and consistent impact on perceived underwater visibility. In both isolated and combined scenes, higher light attenuation distances and reduced bloom intensity shows significantly higher visibility ratings, validating Hypotheses H1 and H2. These effects remained consistent when multiple visual parameters were combined, indicating that visibility is an important parameter in underwater scene perception.

The influence of light color temperature on emotional perception was more subtle when evaluated in isolation. While median values indicated a neutral mood response for both warmer and colder lighting parameters, mean values suggest a weak tendency toward calmer scenes under warmer lighting and tenser scenes under colder lighting. This supports Hypothesis H3 but suggests that color temperature alone has a marginal effect on mood perception. However, in the combined scene OWT 4a, reduced visibility amplified mood responses, indicating that emotional perception is influenced by combination of visibility perception, color temperature and color adjustments.

The combined scene survey data also revealed limitations in accurately depicting real-world OWT. While participants successfully identified the OWT 4a scene in most cases (85%), the lower identification accuracy for the OWT 2 scene (42%) raises questions regarding scene representation, asset selection, reference fidelity and more importantly, the impact of not including other important parameters in underwater scenes like god rays and caustics. Feedback from participants also suggests that the survey methodology, regarding the ways questions are presented, influenced perception and lack of feedback field (e.g. Text box) in the survey to further analyze answers by the participants.

Regarding realism, results indicate less intense visual effects like fog attenuation and bloom were perceived as more realistic than scenes with stronger visual effects. This suggests that while visual enhancements improve clarity and mood differentiation, stronger effects may reduce perceived realism of the scene.

Overall, this study shows that visibility-related parameters play a central role in underwater scene perception, affecting not only visibility but also emotional response and realism. Lighting color temperature and color adjustments contributes to mood perception but is more effective when combined with other visual effects. I believe that this work gives valuable insight to any developer that has little understanding of creating underwater scenes and provides tools and knowledge as a starting point to recreating underwater scenes.

FUTURE WORK

While this work provides valuable insights into the creation of underwater scenes related with realism and mood perception when it comes to manipulation parameters such as volumetric fog attenuation, bloom and particles effects and light color temperature, it does not address other components that are also present in the creation of underwater scenes like caustics and god rays. Due to out-of-scope concerns these parameters were not addressed in this work but were referenced as important factors. Regarding future perspective, this work would benefit substantially with the introduction of god rays and caustics into the scene and the study of their impact together with the parameters already studied in this work.

Furthermore, most engines already have built-in methods to facilitate the recreation of underwater scenes, for example, Unity has Water System, that can create a water surface and automatically create an underwater environment when the player camera is below the surface, showing light attenuation caustics and god rays.

There is also a limitation of this work that needs to be addressed. This work uses a fixed camera that can rotate around the scene, but in games, the player moves in the scene. This represents a big limitation in this work regarding underwater scene representation. First, we need to address how volumetric fog behaves; if we had the possibility of moving the camera freely when going deeper the fog would be stronger, but this does not translate well into the real-life behavior of light because with depth the light attenuation is higher and consequently less light penetrates in deeper environments. In this case the fog would be stronger, limiting visibility to some extent, instead of reducing the light in the environment. So, we would need to have a light intensity based on depth as well as color adjustments.

This also highlights the limitations of the survey that was conducted. For the previous example discussed of future work, a new type of survey would need to be created and improved since the user would need to be able to experiment and move around in the environment and not rely on images to give feedback.

To conclude, one of the more important aspects of this work is data collection. Since we discuss very subjective aspects like visibility, mood and realism, a survey of some sort is required. The survey conducted for this work has some limitations, like the use of single images. This work would benefit greatly if more images or even videos were available and a feedback field where participants could justify their decision. The way that the images and questions are presented should change maybe by randomizing question order or even creating more sections. Going further, allowing the survey participants to play and interact with the environment would give more valuable information regarding immersion.

CRITICAL REFLECTION

When I started this course, I thought my work would be more related with programming. Yet, the choice for this topic came from a personal interest related to underwater environments, and I feel like this work, which is related with the subjective aspect of game development, is often not dealt in programming and is an important step I would like to study. By leaving my comfort zone and focus more on understanding the physical aspect of light behavior and focus on subjective aspects as mood and realism, often associated with art related fields, I got a deeper understanding of what it is to be a game developer.

The opportunity to work on something that comes purely out of personal interest and motivation, in a field where I had little to no knowledge is very rewarding. It allowed me to explore and learn about what it is to be a researcher and to pursue a topic in an academic way by using scientific methods to gather data and extrapolate conclusions. it gives me tools and confidence when in the future I might face the situation where I need to find solutions to problems I have little to no knowledge.

This work allowed me to better know myself when dealing with tasks with a high degree of freedom, where I needed to set my own deadlines and manage my own schedule. It allowed me to improve my time managing and problem-solving skills.

At the end the feeling is bittersweet, for one I am happy with the work I have done, but on the other hand, I see its limitations and have the feeling that in there is more to explore and more to learn.

I believe this course is one of the most important courses in this degree as it helped me the most in developing not only hard skills but also soft skills that I think are going to be very important for anyone in their future career.

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ACKNOWLEDGEMENTS

First, I want to thank my supervisors, Thomas and Stephanie, who helped me and guided me in a very meaningful way. Their continuous support and interest throughout the semester were invaluable.

I also want to thank every participant in the survey that took time out of their busy lives to answer the questions.

And lastly, my wife Rita, not only for her support, but for her valuable insight into the research aspect of this work.

APPENDICES

Table A 1 OWT classification table[8].

OWT	Description
1	Extremely clear and oligotrophic indigo-blue waters with high reflectance in the short visible wavelengths.
2	Blue waters, slightly higher detritus and CDOM content than OWT 1.
3a	Turquoise waters, higher phytoplankton, detritus, and CDOM than OWT 1 and 2.
3b	Special case of OWT 3a; strong scattering, little absorbing particles (Coccolithophore blooms), bright with ~490 nm peak.
4a	Greenish water in coastal/inland regions, higher biomass, shorter wavelength reflectance depressed by absorption.
4b	Special case of OWT 4a, intense phytoplankton blooms with higher scattering, very bright green color.
5a	Green eutrophic water, significantly higher phytoplankton biomass, bimodal reflectance (peaks at ~560 nm and ~709 nm).
5b	Green hyper-eutrophic water, biomass higher than OWT 5a, plateau-like NIR reflectance (vegetation-like spectrum).
6	Bright brown water, high detritus, high reflectance from scattering.
7	Dark brown/black water, very high CDOM, low reflectance across the visible, dominated by absorption.

```
using UnityEngine;

public class CameraOrbit : MonoBehaviour
{
    [Header("Orbit Target")]
    [SerializeField] private Transform target;

    [Header("Orbit Settings")]
    public float distance = 20f;
    public float height = 5f;
    public float rotationSpeed = 20f;
    public float pitchAngle = 20f;
    public bool autoRotate = true;

    private float currentAngle = 0f;

    private void LateUpdate()
    {
        if (target == null) return;

        // Rotate horizontally
        if (autoRotate)
            currentAngle += rotationSpeed * Time.deltaTime;

        float rad = currentAngle * Mathf.Deg2Rad;
        float x = Mathf.Cos(rad) * distance;
        float z = Mathf.Sin(rad) * distance;

        transform.position = new Vector3(
            target.position.x + x,
            target.position.y + height,
            target.position.z + z
        );

        Vector3 direction = target.position - transform.position;
        float yaw = Mathf.Atan2(direction.x, direction.z) * Mathf.Rad2Deg;

        transform.rotation = Quaternion.Euler(pitchAngle, yaw, 0f);
    }
}
```

Figure A 1 Camera script.